corresponding gap for the bank-level sample is 42%, meaning that 42% of the time a bank has no exposure to the sovereign debt of a potential target country of investment. These gaps are consistent with the idea that the sovereign debt market is far from frictionless. There is room, in other words, for factors such as trust to explain variations in banks' sovereign debt portfolios.

4. Empirical Model

We outline the empirical model and discuss the results of the country-level analysis in Online Appendix B. Our main analysis and contribution of the paper rests on the following bank-level specification:

$$\text{Sovereign Exposure}_{b,h,c,t} = \beta_1 \text{Bank-level Trust Bias}_{bc} + \beta_2 \text{Bank Branches}_{bc} + \beta_3 \gamma_{bt} + \beta_4 \lambda_{hct} + \varepsilon_{bhct}, \quad (1)$$

where $Sovereign\ Exposure_{bhct}$ is a dummy variable for whether or not bank b of home country h has any positive exposure to target country c at time t.

We estimate linear probability models, facilitating interpretation of our coefficients as marginal probabilities. A dummy variable provides several advantages (over a continuous one) in this setting. Because of the consolidated nature of EBA disclosures, we cannot distinguish between bonds purchased at headquarters and at subsidiaries. We therefore consider the extensive margin of sovereign exposures, since strategic decisions such as whether or not a bank should invest in a country are taken at bank headquarters. In addition, there is heterogeneity in sovereign debt valuation methods across disclosures and some flexibility at the bank level in categorizing sovereign exposures as residing on the trading versus banking books, which in turn affects reported values. Such flexibility could lead to self-reporting biases for the continuous variable but is less likely to affect its extensive margin (Kaplow and Shavell 1994). Finally, since we do not observe currency denomination, exchange rate fluctuations can introduce variation in reported sovereign exposures in different currencies in the absence of active investment decisions.[27]

$Bank-level\ Trust\ Bias_{bc}$ is constructed by computing a weighted average of *Country-level Trust Bias* across host countries for each bank–target-country pair (b, c), where the share of host-country (i) branches (in numbers) in the overall branch network

each bank's debt portfolio should be proportional to the share of that sovereign's total debt in the sovereign debt market (Sharpe, 1964). By implication, this would require each bank to have at least some positive exposure to each sovereign in our sample, thus implying an unconditional probability of one.

27. Despite its drawbacks, we consider the continuous exposure variable in robustness checks, employing the log of the nominal values (in million Euros) of sovereign lending reported by banks, and obtain similar results. The same holds when we define the dependent variable as the share of the specific target country in the bank's total sovereign debt portfolio (results available on request).

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of the multinational bank is employed as weights:[28]

$$\text{Bank-level Trust Bias}_{b,c} = \sum_{i=1}^{n} (\text{Weight}_{b,i} \times \text{Country-level Trust Bias}_{i,c}). \quad (2)$$

We define *Country-level Bilateral Trust* as the share of respondents in home country h expressing "a lot of trust" in target country c. These self-reported measures are then adjusted for country fixed effects, because some nationalities may be universally regarded as more trustworthy and respondents of some nationalities may trust all foreigners more. In moving from $Country-level\ Bilateral\ Trust\ to\ Country-level\ Trust\ Bias_{hc}$ (the variable that appears in equation (2)), we follow Guiso et al. (2009), Bloom et al. (2012) and Pursiainen (2022), running a gravity regression of bilateral trust for country pairs:

$$\text{Country-level Bilateral Trust}_{h,c} = \alpha_1 \theta_h + \alpha_2 \vartheta_c + \epsilon_{hc}. \quad (3)$$

Residuals from this regression, after controlling for home-country (θ_h) and target-country (ϑ_c) fixed effects, capture the relative trust bias of home country h toward target country c ($\epsilon_{hc} = Country-level\ Trust\ Bias_{hc}$). For comparability across studies, we borrow this measure directly from Pursiainen (2022).[29]

The identification strategy is depicted in panel A of Figure 1. We exploit the variation across banks (i.e., HSBC vs. RBS) headquartered within the same home country (i.e., UK) facing the same target country (i.e., Austria) at the same point in time because these multinational banks have subsidiaries in different countries (i.e., France and Ireland) and because residents of those host countries have different perceptions of the same target country.[30] Hence, saturating our specification with *home-country* $\times$ *target-country* $\times$ *time* fixed effects (λ_{hct} in equation (1)) absorbs all time-varying country-level variation in our outcome variable. For instance, if banks located in EU countries shift away from UK exposures after Brexit, this would be controlled for in our setting. Thanks to this framework, we can also rule out explanations for observed

28. For instance, if a bank has 50% of its branches in country A and 50% in country B, then its trust bias toward country C is the simple average of trust biases in countries A and B toward country C. Recall that, in line with the notion of persistent cultural stereotypes, this measure is time-invariant and constructed from a single snapshot of bank branch networks for each bank. Hence, we do not have time variation in bank-level trust. That said, changes in branch networks tend to be very gradual. In addition, previous literature has emphasized the long-term stability of cultural stereotypes and used time-invariant measures to capture them (see, e.g., Guiso et al., 2009; Bloom et al., 2012; Bottazzi et al., 2016; Pursiainen, 2022). As we show below, our new survey from 2022 provides further evidence on the persistence of cultural stereotypes over time.

29. The resulting measure is illustrated in Online Appendix Table A.5 (see Table IA.II in Pursiainen (2022)) for Eurobarometer and Online Appendix Table D.3 for our online survey, whereas the corresponding Country-level Bilateral Trust (in levels without the gravity adjustment in equation 3) is reported in Online Appendix Table A.4 (see Table IA.I in Pursiainen (2022)) and Online Appendix Table D.2.

30. Domestic banks (i.e., Lloyds), on the other hand, do not add to our identifying variation as their treatment status only depends on the variation between home and target countries.

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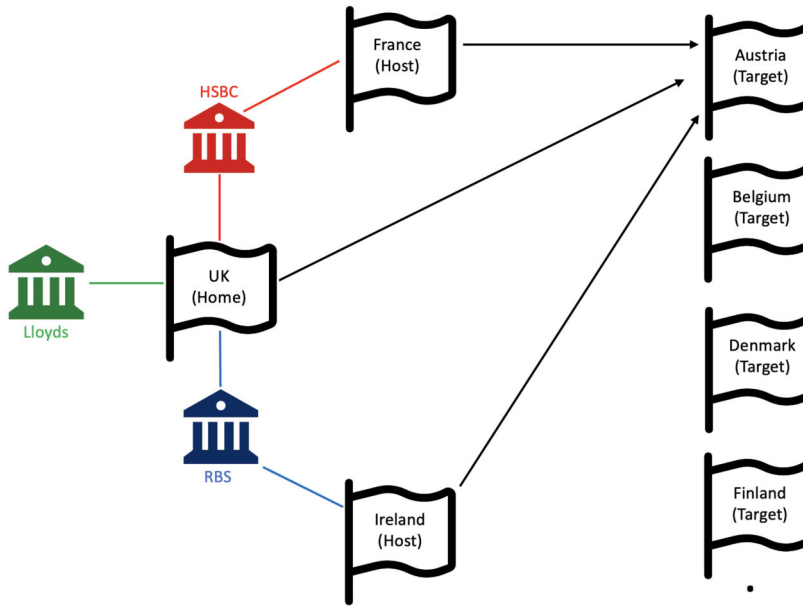
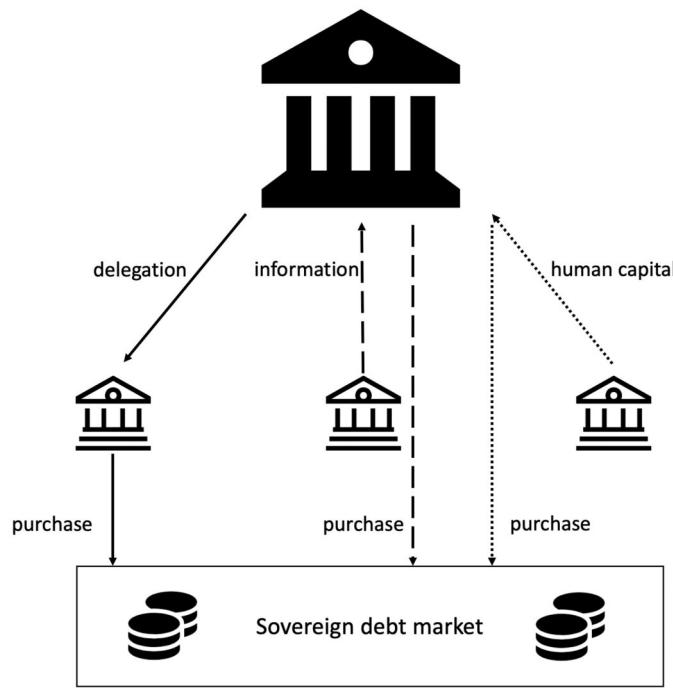


(a) Identification strategy at bank level



(b) Banks as hierarchies

FIGURE 1. Bank-level identification strategy and framework. This figure represents (a) the bank-level identification strategy as described in Section 4 and (b) the mechanisms that link foreign bank branches to multinational banks' sovereign exposures as described in Section 6.

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investment behavior that attributes differences across banks to differential exposure to default risk, since it is unlikely that a sovereign will default on the bonds held by one bank but not another headquartered in the same foreign country.[31]

Banks hold sovereign bonds for different reasons. Some hold them for trading with clients (Duffie 2010). Others hold bonds of risky sovereigns because they are confident of being bailed out by the authorities (Admati and Hellwig 2013). Still others purchase government bonds as the price of obtaining other advisory, cash-management or investment-management mandates.[32] Insofar as motives and behaviors differ across banks, bank fixed effects will pick them up. Insofar as they differ over time, we can control for such variation by including *bank* × *time* fixed effects (γ_{bt} in equation (1)). We consequently exploit the variation across different target countries for the same bank at the same point in time, shielding our estimates from time-varying omitted shocks operating at the bank level.

We also control for the number of branches a bank has in a target country ($Bank\,Branches_{bc}$). This helps to distinguish the information channel (and, more broadly, direct financial linkages between banks and target countries) highlighted in Saka (2020). In additional analyses, we compare banks' lending to target countries in which none of the banks considered have branches (e.g., in Figure 1a, dropping observations of investments in the United Kingdom, France, and Ireland when comparing RBS and HSBC). We control for indirect relationships between banks and target countries that may be sustained through host countries (e.g., HSBC—compared to RBS—being financially closer to Austria because France is financially better linked to Austria than is Ireland). Finally, we control for various measures of cultural distance between banks and target countries to show that the role of trust cannot be fully explained by cultural differences.

We cluster standard errors by bank, given the possibility that the error term is correlated across target countries and time. Double clustering at country-pair and time levels or double clustering at country-pair and bank levels does not change the results.

31. We know of no 21st-century European case (that being the place and period of our data) where one class of foreign banks was treated better than another. Looking more broadly, we can find cases where some classes of nonbank creditors were treated more favorably (hedge funds that held out in the Argentine case). One can also find cases outside Europe where a government defaulted on external bonds but not on domestic bonds (in order to protect its banking system) and cases (such as Jamaica) where a government restructured bonds held by domestic banks but not foreign banks (to protect its foreign capital market access). Importantly, our results hold when we exclude domestic banks of the target country of investment from the sample. Above all, our results hold when we use fixed effects to limit our comparisons to banks headquartered in the same country lending to the same target country at the same point in time. So, for instance, it would have to be the Greek government treating Deutsche Bank and Commerzbank differentially, not treating German and Spanish banks differentially. We know of no evidence or discussion of such differential within-country treatment.

32. On competition for mandates more generally, see Ljungqvist, Marston, and Wilhelm (2006).

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